

Fat7 Links

D. A. Clarke (HotQCD, Fermilab-MILC)

The Fat7-smeared links are [1]

$$V_\mu = c_1 U_\mu + \sum_{\nu \neq \mu} \left[c_3 U_{\mu\nu}^{(3)} + \sum_{\substack{\rho \neq \mu \\ \rho \neq \nu}} \left(c_5 U_{\mu\nu\rho}^{(5)} + \sum_{\substack{\sigma \neq \mu \\ \sigma \neq \nu \\ \sigma \neq \rho}} c_7 U_{\mu\nu\rho\sigma}^{(7)} \right) \right], \quad (1)$$

where summations run over both forward and backward directions, and the link constructs are given by

$$\begin{aligned} U_{\mu\nu}^{(3)}(x) &= U_\nu(x) U_\mu(x + a\hat{\nu}) U_\nu^\dagger(x + a\hat{\mu}), \\ U_{\mu\nu\rho}^{(5)}(x) &= U_\nu(x) U_{\mu\rho}^{(3)}(x + a\hat{\nu}) U_\nu^\dagger(x + a\hat{\mu}), \\ U_{\mu\nu\rho\sigma}^{(7)}(x) &= U_\nu(x) U_{\mu\rho\sigma}^{(5)}(x + a\hat{\nu}) U_\nu^\dagger(x + a\hat{\mu}). \end{aligned} \quad (2)$$

The typical choice for the coefficients is

$$c_1 = \frac{1}{8}, \quad c_3 = \frac{1}{16}, \quad c_5 = \frac{1}{64}, \quad c_7 = \frac{1}{384}. \quad (3)$$

References

- [1] K. Orginos, D. Toussaint, and R. L. Sugar. Variants of fattening and flavor symmetry restoration. *Phys. Rev. D*, 60(5):054503, 1999.